

Image Processing & Computer Vision

Informatik
Prof. Dr. Thomas Koller
Dozent

thomas.koller@hslu.ch



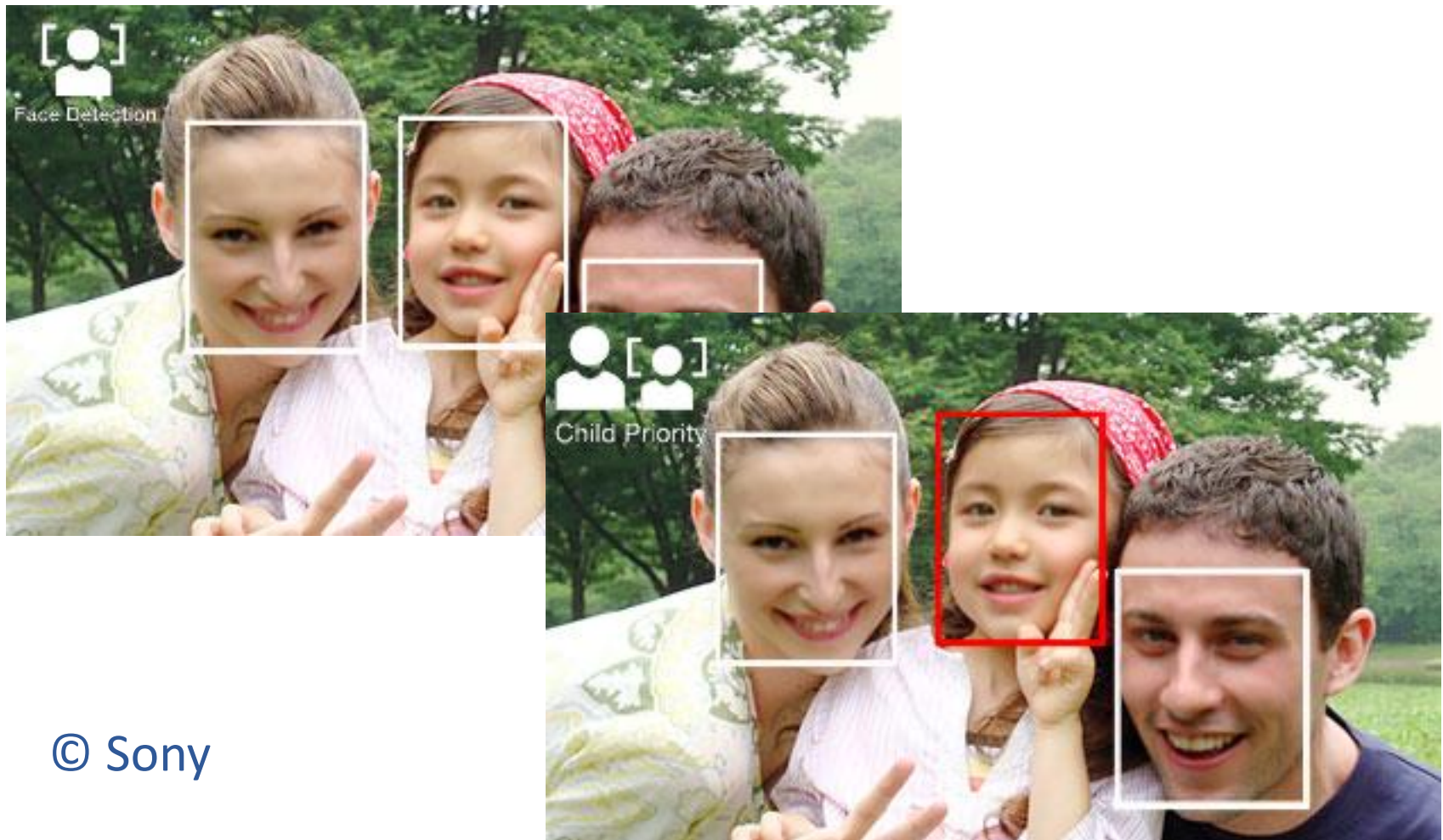
Willkommen

- Wieso sind Sie hier?
- Was sind Ihre Erwartungen?

Motivation

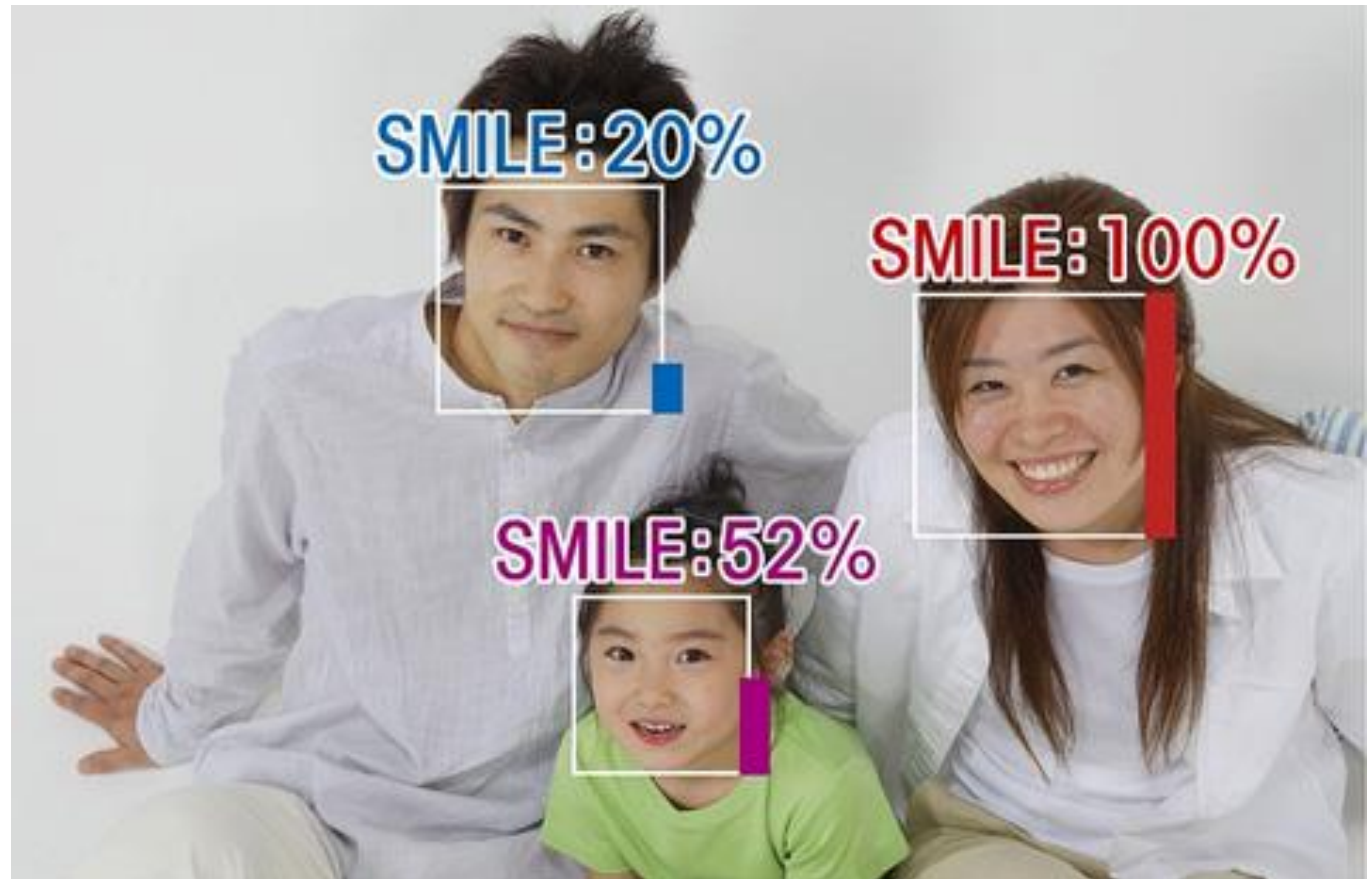
- Wo braucht es Computer Vision?

Face Detection



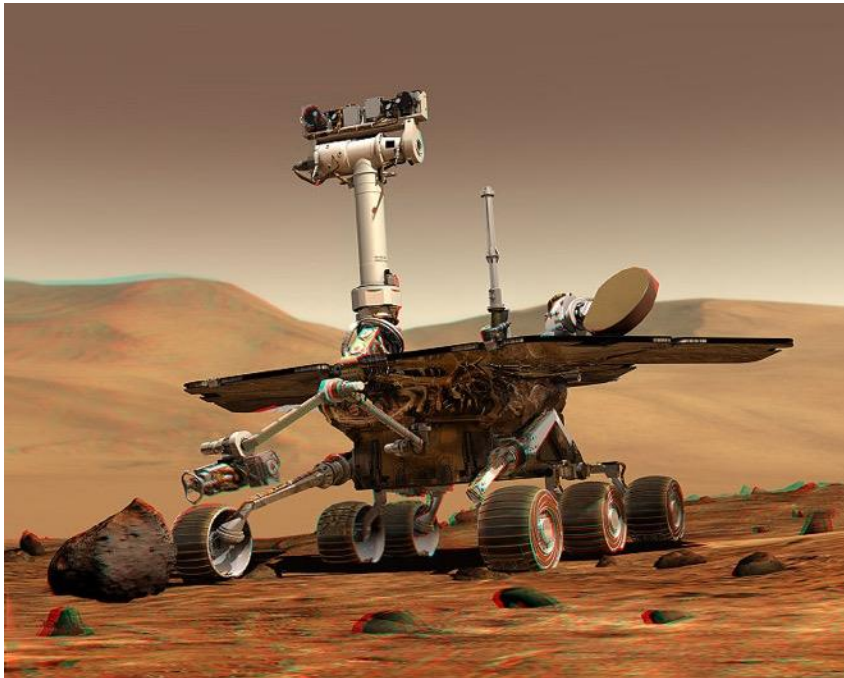
© Sony

Well...



©Akibahara News

Robotics



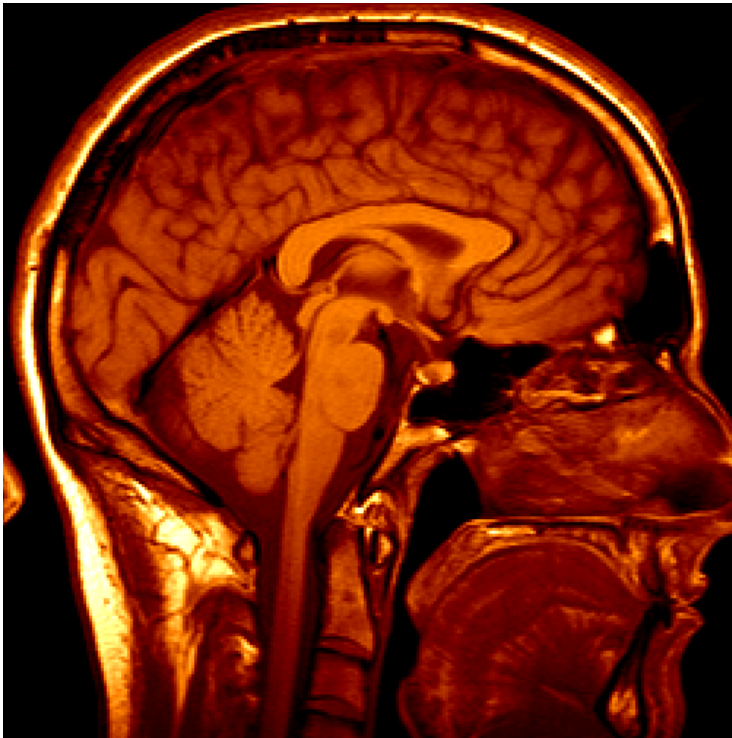
NASA's Mars Spirit Rover

http://en.wikipedia.org/wiki/Spirit_rover



<http://www.robocup.org/>

Medical Imaging



3D imaging
MRI, CT



Image guided surgery
[Grimson et al., MIT](#)

Medical imaging

**Joseph Stiglitz on
artificial intelligence:
'We're going towards a
more divided society'**

<https://www.theguardian.com/technology/2018/sep/08/joseph-stiglitz-on-artificial-intelligence-were-going-towards-a-more-divided-society>



▲ Microsoft's InnerEye project uses AI to make treatment for prostate cancer more efficient.
Photograph: Microsoft Project InnerEye Study



Google Car (jetzt Waymo)

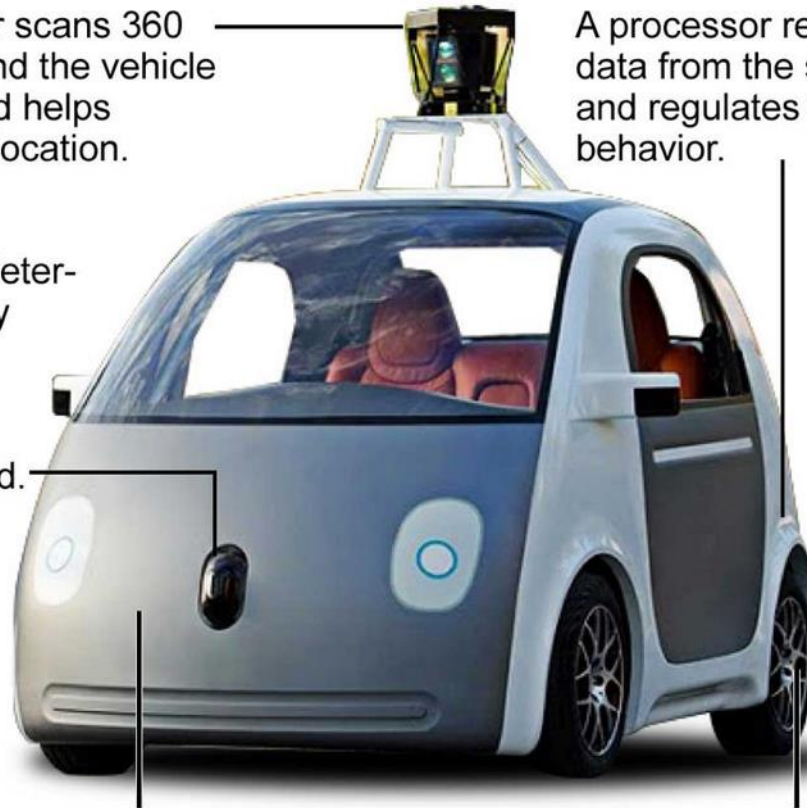
A laser sensor scans 360 degrees around the vehicle for objects and helps determine its location.

A processor reads the data from the sensors and regulates vehicle behavior.

Radar helps determine speed by detecting and measuring the speed of vehicles ahead.

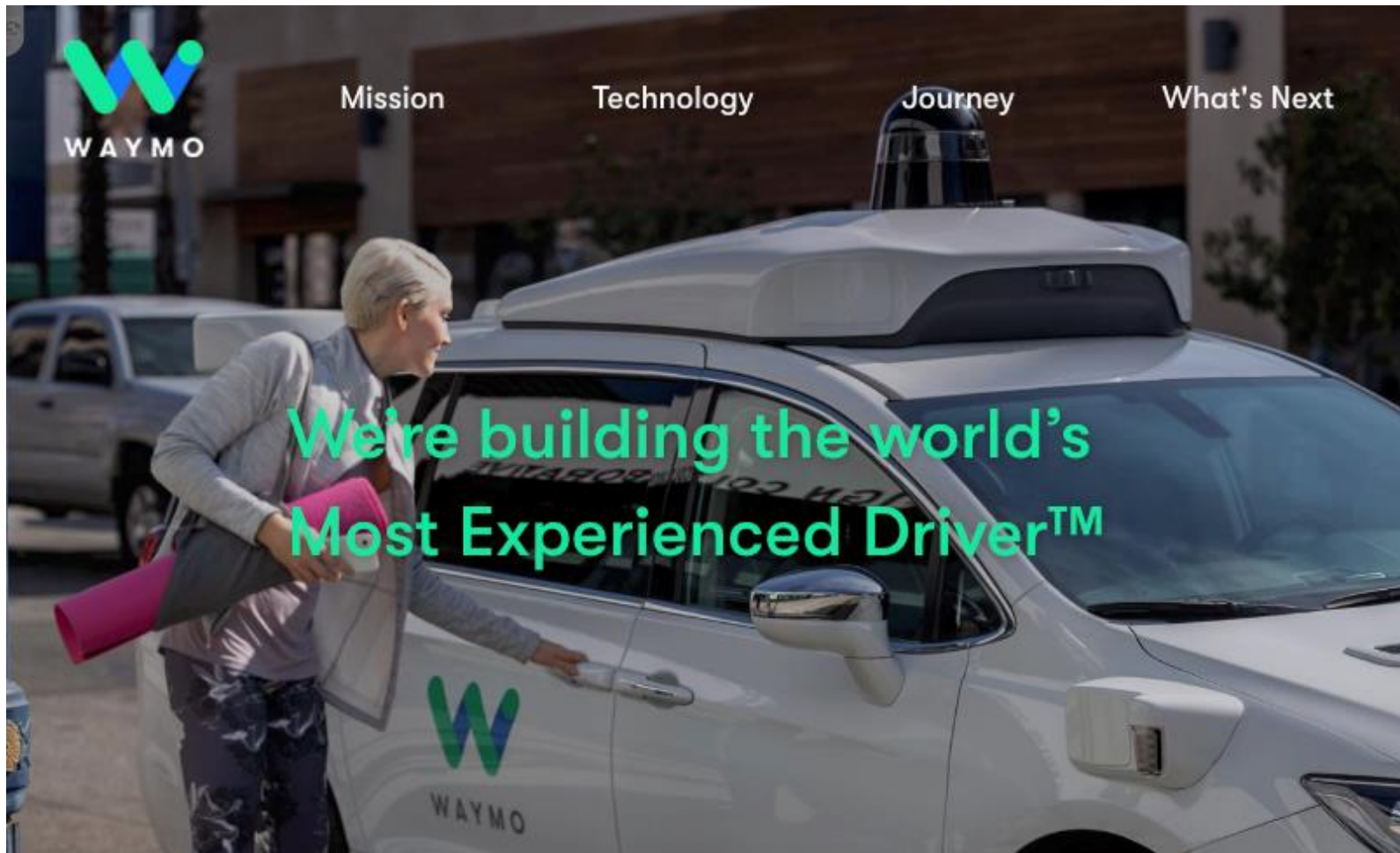
Orientation sensor located inside the car tracks the car's motion and balance.

Wheel hub sensor detects the number of rotations to help determine the car's location.

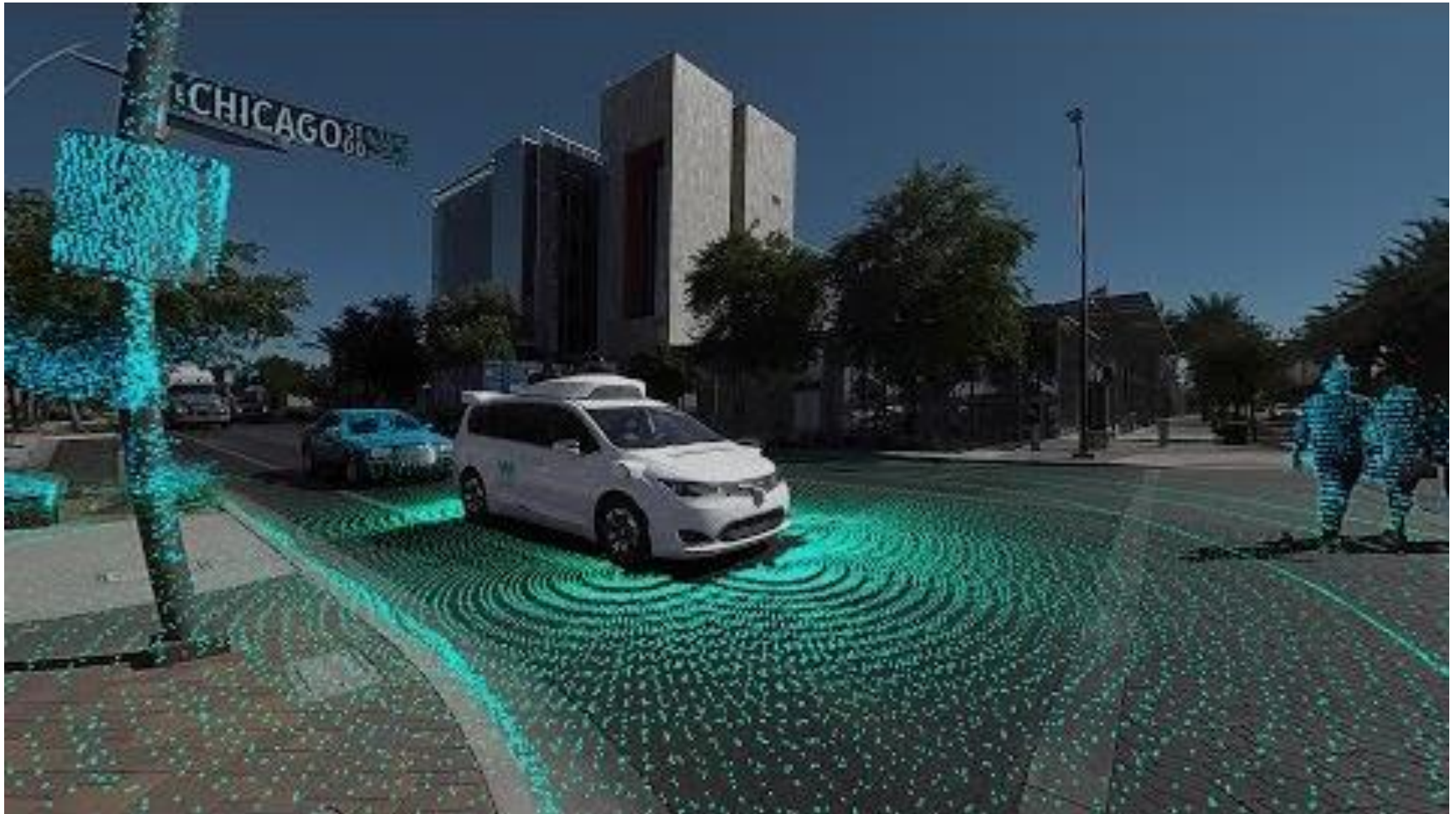


<http://www.latimes.com/>

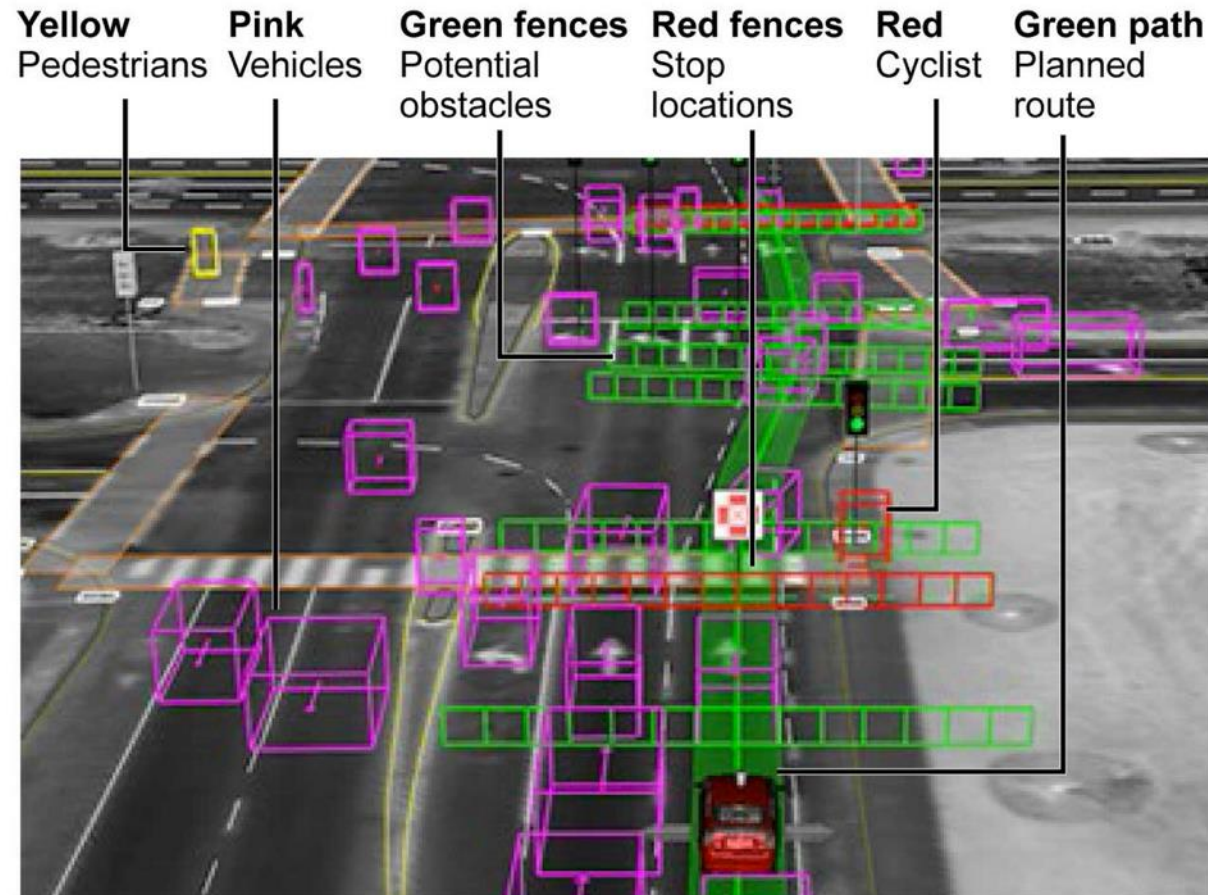
Waymo



Waymo



How the car sees the world

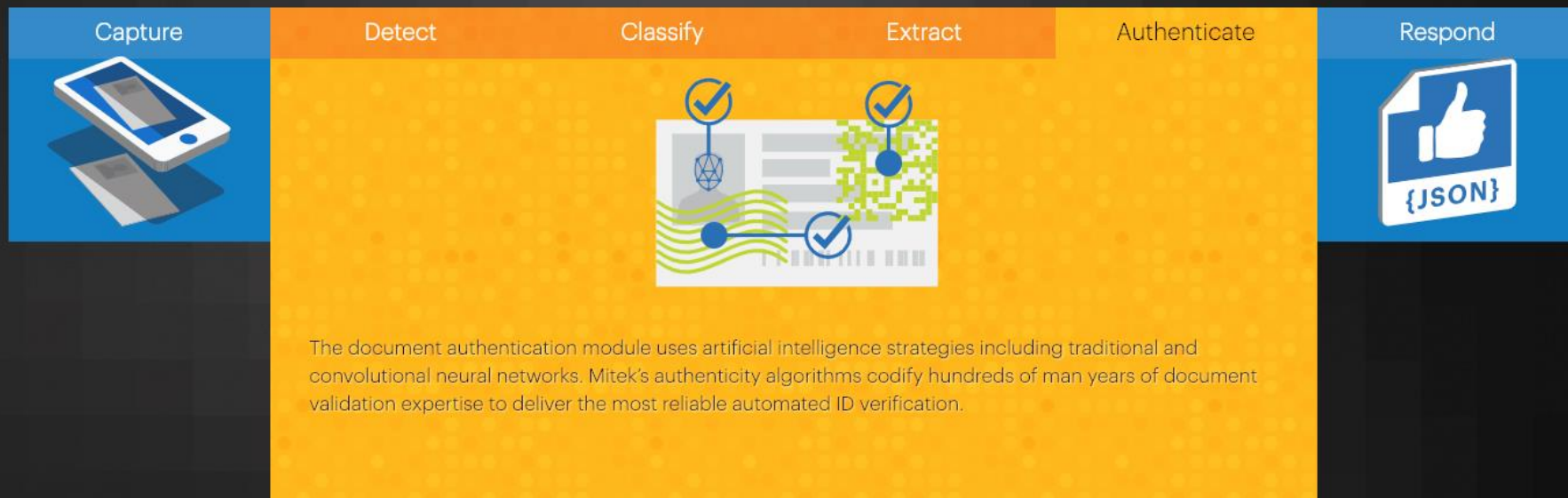


<https://waymo.com/tech/>

Banking

Core Flow™ - World's Most Powerful Mobile Capture & ID Verification Engine

Built on patented algorithms, Mitek's Core Flow™ engine is a modular cross – platform architecture which utilizes machine learning to classify, extract data, and authenticate documents including passports, ID cards, driver's licenses, and checks.



<https://www.miteksystems.com/technology>

Entertainment



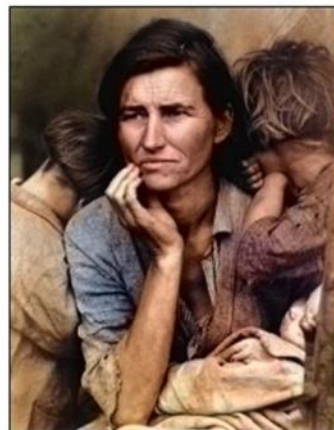
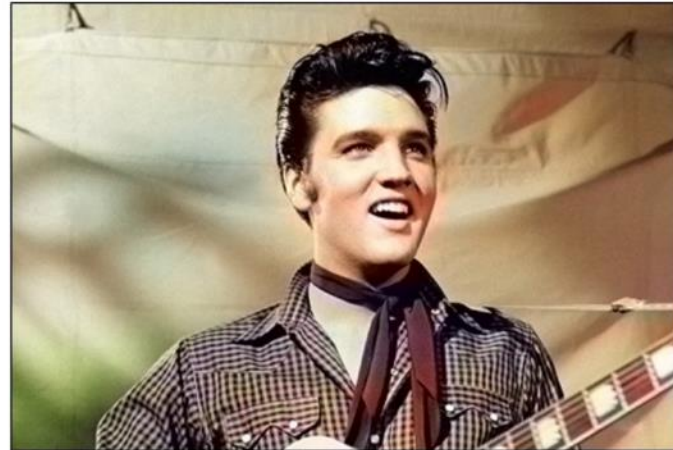
<http://www.ilm.com/>

Image Processing & Computer Vision

Shape Capture

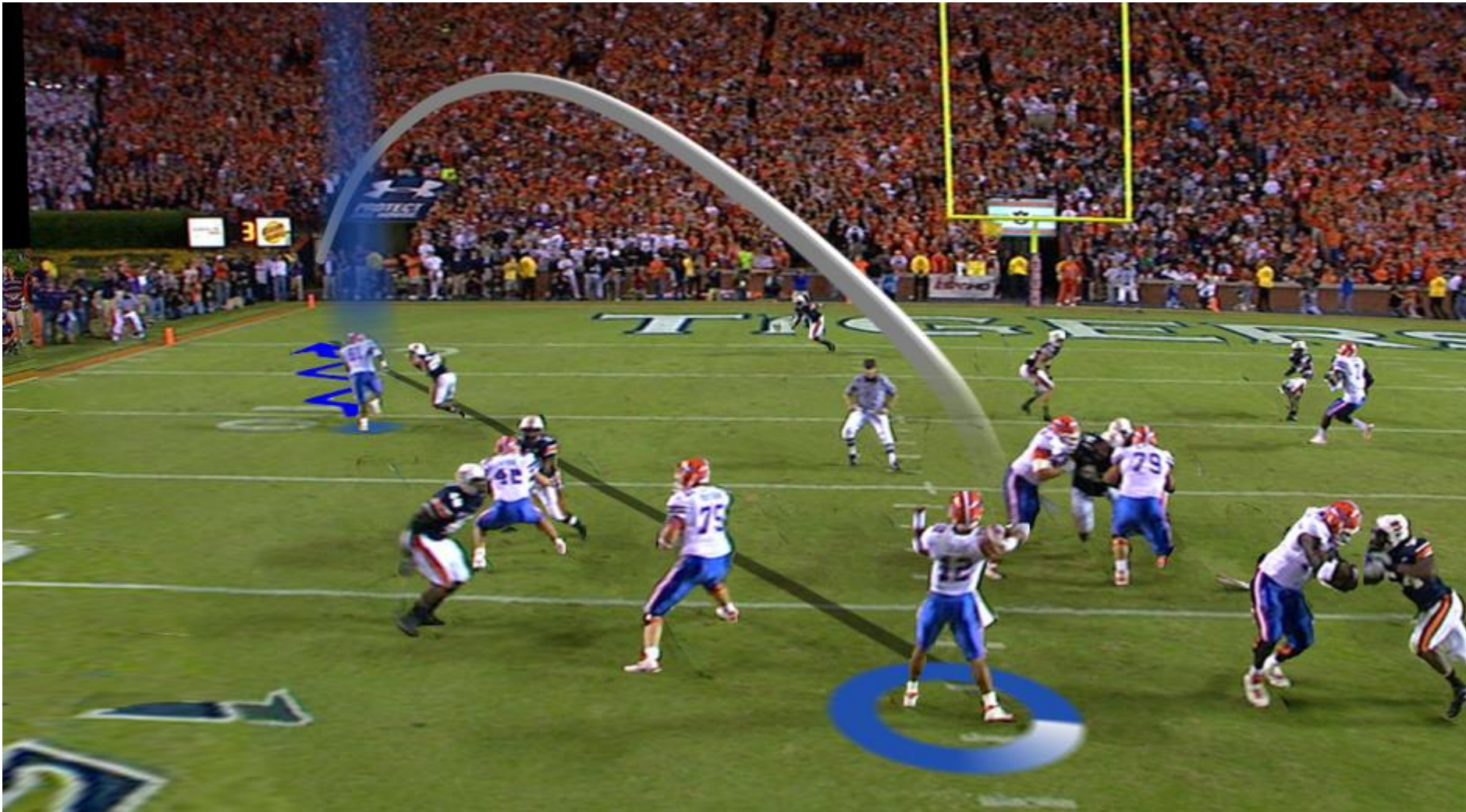


Colorization



Colorful Image Colorization, Zhang et. al., University of Berkeley

Broadcasting

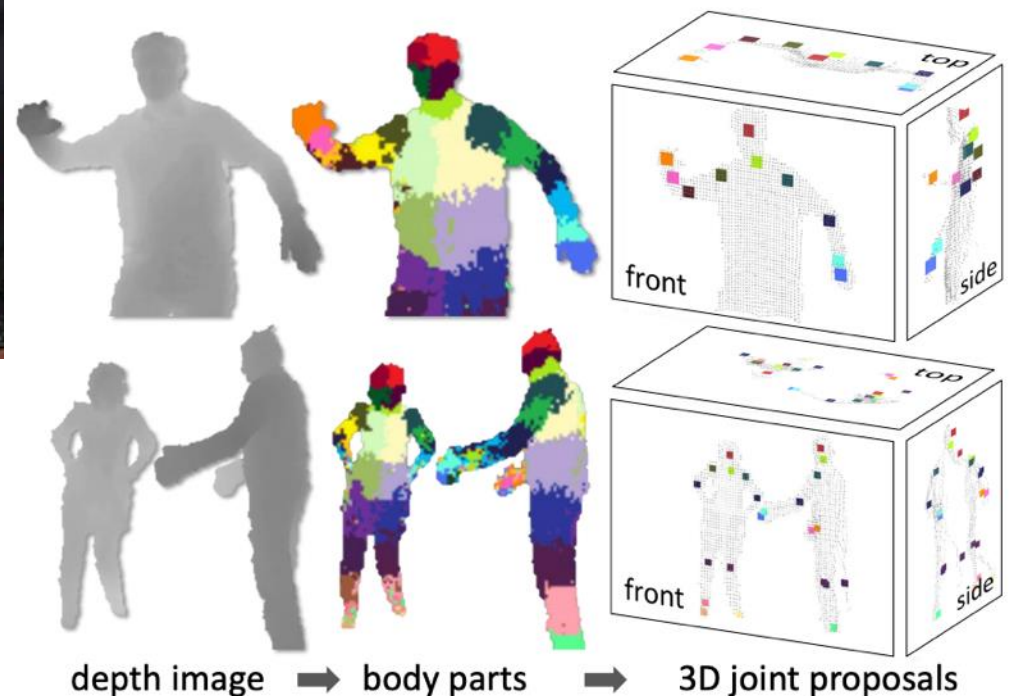


Kinect Pose Recognition

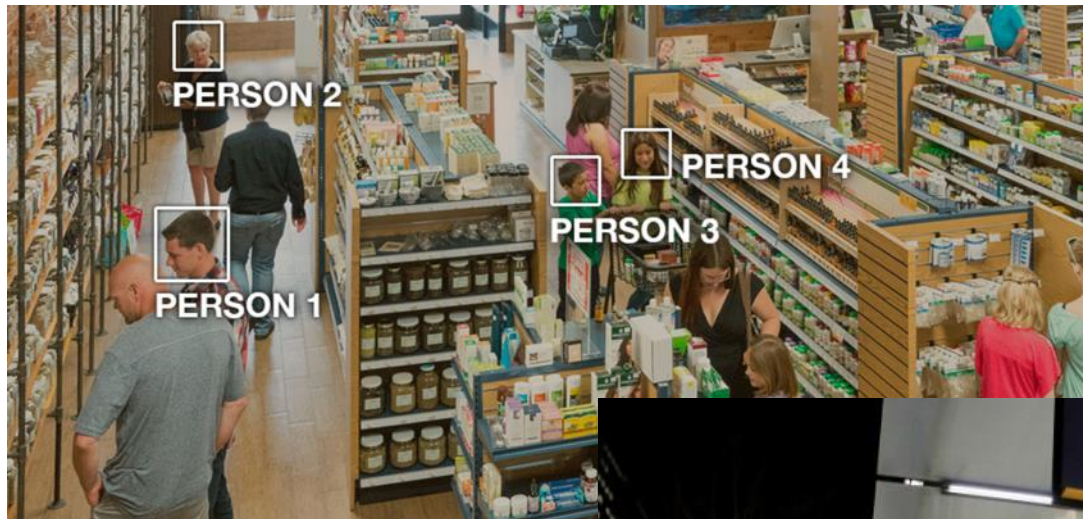


Researchers from Microsoft Research Cambridge beam during the ceremony in which they won the MacRobert Award: (from left) Mat Cook, Jamie Shotton, Andrew Blake, Andrew Fitzgibbon, and Toby Sharp.

Real-Time Human Pose Recognition in Parts from Single Depth Images



Retail



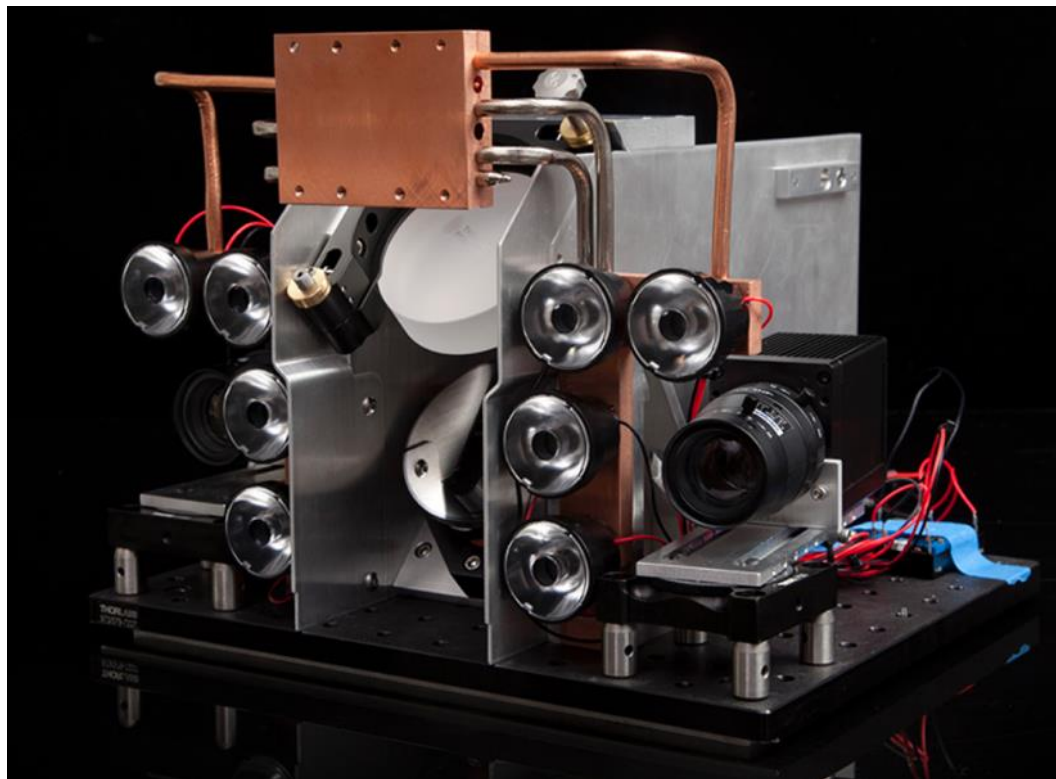
Smarter Farming



SlantRange Advanced Crop Information Solutions



Photonic-Fence Prototyp



Killing
mosquitos 😊



INTELLECTUAL VENTURES
LABORATORY

Fakes...

The Washington Post
Democracy Dies in Darkness

A Seattle TV station aired doctored footage of Trump's Oval Office speech. The employee has been fired.



